

Multiplex Enhancer Interference Reveals Collaborative Control of Gene Regulation by Estrogen Receptor α -Bound Enhancers.

PMID: 28964699 | Indexed in PubMed

Multiple regulatory regions have the potential to regulate a single gene, yet how these elements combine to affect gene expression remains unclear. To uncover the combinatorial relationships between enhancers, we developed Enhancer-interference (Enhancer-i), a CRISPR interference-based approach that uses 2 different repressive domains, KRAB and SID, to prevent enhancer activation simultaneously at multiple regulatory regions. We applied Enhancer-i to promoter-distal estrogen receptor α binding sites (ERBS), which cluster around estradiol-responsive genes and therefore may collaborate to regulate gene expression. Targeting individual sites revealed predominant ERBS that are completely required for the transcriptional response, indicating a lack of redundancy. Simultaneous interference of different ERBS combinations identified supportive ERBS that contribute only when predominant sites are active. Using mathematical modeling, we find strong evidence for collaboration between predominant and supportive ERBS. Overall, our findings expose a complex functional hierarchy of enhancers, where multiple loci bound by the same transcription factor combine to fine-tune the expression of target genes.

Enhancer modularity and the evolution of new traits.

PMID: 26925592 | Indexed in PubMed

Animals have modular cis-regulatory regions in their genomes, and expression of a single gene is often regulated by multiple enhancers residing in such a region. In the laboratory, and also in natural populations, loss of an enhancer can result in a loss of gene expression. Although only a few examples have been well characterized to date, some studies have suggested that an evolutionary gain of a new enhancer function can establish a new gene expression domain. Our recent study showed that *Drosophila guttifera* has more enhancers and additional expression domains of the wingless gene during the pupal stage, compared to *D. melanogaster*, and that these new features appear to have evolved in the ancestral lineage leading to *D. guttifera*. (1) Gain of a new expression domain of a developmental regulatory gene (toolkit gene), such as wingless, can cause co-option of the expression of its downstream genes to the new domain, resulting in duplication of a preexisting structure at this new body position. Recently, with the advancement of evo-devo studies, we have learned that the developmental regulatory systems are strikingly similar across various animal taxa, in spite of the great diversity of the animals' morphology. Even behind "new" traits, co-options of essential developmental genes from known systems are very common. We previously provided concrete evidence of gains of enhancer activities of a developmental regulatory gene underlying gains of new traits. (1) Broad occurrence of this scenario is testable and should be validated in the future.

Enhancer diversity and the control of a simple pattern of *Drosophila* CNS midline cell expression.

PMID: 24854999 | Indexed in PubMed

Transcriptional enhancers integrate information derived from transcription factor binding to control gene expression. One key question concerns the extent of trans- and cis-regulatory variation in how

co-expressed genes are controlled. The *Drosophila* CNS midline cells constitute a group of neurons and glia in which expression changes can be readily characterized during specification and differentiation. Using a transgenic approach, we compare the cis-regulation of multiple genes expressed in the *Drosophila* CNS midline primordium cells, and show that while the expression patterns may appear alike, the target genes are not equivalent in how these common expression patterns are achieved. Some genes utilize a single enhancer that promotes expression in all midline cells, while others utilize multiple enhancers with distinct spatial, temporal, and quantitative contributions. Two regulators, Single-minded and Notch, play key roles in controlling early midline gene expression. While Single-minded is expected to control expression of most, if not all, midline primordium-expressed genes, the role of Notch in directly controlling midline transcription is unknown. Midline primordium expression of the rhomboid gene is dependent on cell signaling by the Notch signaling pathway. Mutational analysis of a rhomboid enhancer reveals at least 5 distinct types of functional cis-control elements, including a binding site for the Notch effector, Suppressor of Hairless. The results suggest a model in which Notch/Suppressor of Hairless levels are insufficient to activate rhomboid expression by itself, but does so in conjunction with additional factors, some of which, including Single-minded, provide midline specificity to Notch activation. Similarly, a midline glial enhancer from the *argos* gene, which is dependent on EGF/Spitz signaling, is directly regulated by contributions from both Pointed, the EGF transcriptional effector, and Single-minded. In contrast, midline primordium expression of other genes shows a strong dependence on Single-minded and varying combinations of additional transcription factors. Thus, Single-minded directly regulates midline primordium-expressed genes, but in some cases plays a primary role in directing target gene midline expression, and in others provides midline specificity to cell signaling inputs.

Transcriptional Dysregulation of MYC Reveals Common Enhancer-Docking Mechanism.

PMID: 29641996 | Indexed in PubMed

Transcriptional dysregulation of the MYC oncogene is among the most frequent events in aggressive tumor cells, and this is generally accomplished by acquisition of a super-enhancer somewhere within the 2.8 Mb TAD where MYC resides. We find that these diverse cancer-specific super-enhancers, differing in size and location, interact with the MYC gene through a common and conserved CTCF binding site located 2 kb upstream of the MYC promoter. Genetic perturbation of this enhancer-docking site in tumor cells reduces CTCF binding, super-enhancer interaction, MYC gene expression, and cell proliferation. CTCF binding is highly sensitive to DNA methylation, and this enhancer-docking site, which is hypomethylated in diverse cancers, can be inactivated through epigenetic editing with dCas9-DNMT. Similar enhancer-docking sites occur at other genes, including genes with prominent roles in multiple cancers, suggesting a mechanism by which tumor cell oncogenes can generally hijack enhancers. These results provide insights into mechanisms that allow a single target gene to be regulated by diverse enhancer elements in different cell types.

Functional characterization of QT interval associated SCN5A enhancer variants identify combined additive effects.

PMID: 39354714 | Indexed in PubMed

Several empirical and theoretical studies suggest the presence of multiple enhancers per gene that

collectively regulate gene expression, and that common sequence variation impacting on the activities of these enhancers is a major source of inter-individual gene expression variability. However, for the vast majority of genes, enhancers and the underlying regulatory variation remains unknown. Even for the genes with well-characterized enhancers, the nature of the combined effects from multiple enhancers and their variants, when known, on gene expression regulation remains unexplored. Here, we have evaluated the combined effects from five SCN5A enhancers and their regulatory variants that are known to collectively correlate with SCN5A cardiac expression and underlie QT interval association in the general population. Using small deletions centered at the regulatory variants in episomal reporter assays in a mouse cardiomyocyte cell line, we demonstrate that the variants and their flanking sequences play critical role in individual enhancer activities, likely being a transcription factor (TF) binding site. By oligonucleotide-based pulldown assays on predicted TFs, we identify the TFs likely driving allele-specific enhancer activities. Using all 32 possible allelic synthetic constructs in reporter assays, representing the five bi-allelic enhancers, we demonstrate combined additive effects on overall enhancer activities. Using transient enhancer assays in zebrafish embryos we demonstrate that four elements act as enhancers in vivo. Together, these studies uncover the TFs driving the enhancer activities of QT interval associated SCN5A regulatory variants, reveal the additive effects from allelic combinations of these regulatory variants, and prove their potential to act as enhancers in vivo.

Regulatory sharing between estrogen receptor α bound enhancers.

PMID: 32479598 | Indexed in PubMed

The human genome encodes an order of magnitude more gene expression enhancers than promoters, suggesting that most genes are regulated by the combined action of multiple enhancers. We have previously shown that neighboring estrogen-responsive enhancers exhibit complex synergistic contributions to the production of an estrogenic transcriptional response. Here we sought to determine the molecular underpinnings of this enhancer cooperativity. We generated genetic deletions of four estrogen receptor α (ER) bound enhancers that regulate two genes and found that enhancers containing full estrogen response element (ERE) motifs control ER binding at neighboring sites, while enhancers with pre-existing histone acetylation/accessibility confer a permissible chromatin environment to the neighboring enhancers. Genome engineering revealed that two enhancers with half EREs could not compensate for the lack of a full ERE site within the cluster. In contrast, two enhancers with full EREs produced a transcriptional response greater than the wild-type locus. By swapping genomic sequences, we found that the genomic location of a full ERE strongly influences enhancer activity. Our results lead to a model in which a full ERE is required for ER recruitment, but the presence of a pre-existing permissible chromatin environment can also be needed for estrogen-driven gene regulation to occur.

Gene regulation: enhancers stepping out of the shadow.

PMID: 20833307 | Indexed in PubMed

The expression of many animal genes has been shown to be controlled by two--rather than one--enhancers with similar regulatory content. Such enhancer redundancy ensures robustness of gene expression under adverse environmental or genetic conditions.

Single-allele chromatin interactions identify regulatory hubs in dynamic compartmentalized domains.

PMID: 30374068 | Indexed in PubMed

The promoters of mammalian genes are commonly regulated by multiple distal enhancers, which physically interact within discrete chromatin domains. How such domains form and how the regulatory elements within them interact in single cells is not understood. To address this we developed Tri-C, a new chromosome conformation capture (3C) approach, to characterize concurrent chromatin interactions at individual alleles. Analysis by Tri-C identifies heterogeneous patterns of single-allele interactions between CTCF boundary elements, indicating that the formation of chromatin domains likely results from a dynamic process. Within these domains, we observe specific higher-order structures that involve simultaneous interactions between multiple enhancers and promoters. Such regulatory hubs provide a structural basis for understanding how multiple cis-regulatory elements act together to establish robust regulation of gene expression.

Towards genome-wide prediction and characterization of enhancers in plants.

PMID: 27321818 | Indexed in PubMed

Enhancers are important cis-regulatory DNA elements that regulate transcription programs by recruiting transcription factors and directing them to the promoters of target genes in a cell-type/tissue-specific manner. The expression of a gene can be regulated by one or multiple enhancers at different developmental stages and/or in different tissues. Enhancers are difficult to identify because of their unpredictable positions relative to their cognate promoters. Remarkably, only a handful of enhancers have been identified in plant species largely due to the lack of general approaches for enhancer identification. Extensive genomic and epigenomic research in mammalian species has revealed that the genomic locations of enhancers can be predicted based on the binding sites of transcriptional co-factors and several distinct features associated with open chromatin. Here we review the methodologies used in enhancer prediction in mammalian species. We also review the recent applications of these methodologies in *Arabidopsis thaliana* and discuss the future directions of enhancer identification in plants. This article is part of a Special Issue entitled: Plant Gene Regulatory Mechanisms and Networks, edited by Dr. Erich Grotewold and Dr. Nathan Springer.

Enhancer-mediated reporter gene expression in *Arabidopsis thaliana*: a forward genetic screen.

PMID: 33547831 | Indexed in PubMed

Gene expression is controlled and regulated by interactions between cis-regulatory DNA elements (CREs) and regulatory proteins. Enhancers are one of the most important classes of CREs in eukaryotes. Eukaryotic genes, especially those related to development or responses to environmental cues, are often regulated by multiple enhancers in different tissues and/or at different developmental stages. Remarkably, little is known about the molecular mechanisms by which enhancers regulate gene expression in plants. We identified a distal enhancer, CRE β , which regulates the expression of AtDGK7, which encodes a diacylglycerol kinase in *Arabidopsis*. We developed a transgenic line containing the

luciferase reporter gene (LUC) driven by CRE β fused with a minimal cauliflower mosaic virus (CaMV) 35S promoter. The CRE β enhancer was shown to play a role in the response to osmotic pressure of the LUC reporter gene. A forward genetic screen pipeline based on the transgenic line was established to generate mutations associated with altered expression of the LUC reporter gene. We identified a suite of mutants with variable LUC expression levels as well as different segregation patterns of the mutations in populations. We demonstrate that this pipeline will allow us to identify trans-regulatory factors associated with CRE β function as well as those acting in the regulation of the endogenous AtDGK7 gene.
